MUSICAL PLAYLIST MANAGER

Project Report

Language: C++
Data Structure: Doubly Linked List
Project Type: Console-based application

1. Introduction

A playlist contains songs that users may browse in sequence. A Doubly Linked List is suitable for this task because each node links to both the previous and next song. This project demonstrates dynamic memory allocation and pointer-based list operations.

2. Problem Statement

Develop a menu-driven program to add, display, search, delete, navigate, reverse, and clear songs in a playlist. Each song stores a title, artist, and album.

3. Objectives

• Implement a Doubly Linked List in C++.
• Maintain `head`, `tail`, and `current` pointers.
• Support forward and backward navigation.
• Perform search and deletion by song title.
• Reverse the list by swapping node links.
• Free dynamically allocated memory.

4. Data Structure

Each `Song` node stores three strings—title, artist, and album—and two pointers: `prev` and `next`. The `head` points to the first node, `tail` to the last node, and `current` identifies the selected song.

5. Algorithms / Working

Add: Create a node and attach it after the tail. If the list is empty, set head, tail, and current to the new node.
Display: Traverse from head using next pointers.
Next / Previous: Move current through next or prev when a neighbour exists.
Search: Traverse the list and compare each title with the requested title.
Delete: Update neighbouring links and head/tail when needed, then delete the node.
Reverse: Swap prev and next for every node, then swap head and tail.
Clear: Delete every node and reset all pointers.

6. Complexity Analysis

Operation	Typical time complexity
Add song at end	O(1)
Display playlist	O(n)
Search by title	O(n)

Delete by title	O(n)
Reverse playlist	O(n)
Clear playlist	O(n)

7. Testing

Test the application by adding multiple songs, displaying them, navigating forward and backward, searching for an existing and a missing title, deleting first/middle/last songs, reversing the playlist, clearing it, and trying operations on an empty playlist.

8. Limitations

The application manages song information only. It does not connect to audio files or play actual music. Song-title matching is case-sensitive and requires an exact title.

9. Conclusion

The Musical Playlist Manager demonstrates practical use of a Doubly Linked List in C++. It provides song management and two-way navigation while reinforcing pointer manipulation, dynamic memory allocation, and list traversal.